

$a \Rightarrow b$

$$N_i \cdot N_j = Qm_i \cdot Qm_j = (Qm_i)^T \cdot (Qm_j) = m_i^T \cdot Q^T \cdot Q \cdot m_j = m_i^T \cdot m_j = m_i \cdot m_j$$

Pro $\forall i, j$ prosto plati, $\tilde{G}_{1,i,j} = G_{2,i,j}$, $\text{tedy } G_1 = G_2$.

$a \Leftarrow b$

Uvažujme matici $Q = [id]_{B_2}^{B_1}$. Pro ni plati $Qm_i = n_i \quad \forall i$.

$$Q = ([m_1]_{B_2} \mid [m_2]_{B_2} \mid \dots \mid [m_m]_{B_2})$$